

Vedāntic Pañcikaṛaṇa and the Emergence of Collective Neural Dynamics via Ephaptic Coupling

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Abstract

In this note the authors provide a remarkable parallel of pañcikaṛaṇa from neuroscience suggesting that the neuroscience phenomenon is downstream from the Vedāntic construction of the gross manifestation.

1 Introduction

This work reveals a profound structural parallelism between the Pañcikaṛaṇa (“quintuplication” or grossification) doctrine of creation in Advaita Vedānta, as expounded by Ādi Śaṅkarācārya¹ in *Tattva Bodha*, and the mathematics of ephaptic coupling in large axonal tracts. In Vedānta, the five pure subtle elements (*tanmātras*) cannot directly produce a gross physical body; they must first undergo a precise weighted mixing process whereby each gross element (*mahābhūta*) consists of 50% of its own subtle element and exactly 12.5% of each of the other four. Only this interdependent blending permits the manifestation of the observable world.

An analogous principle governs ephaptic coupling: isolated axonal membranes cannot give rise to rapid, tract-wide synchrony. Functional collective dynamics emerge only when every axon receives a weighted contribution from the extracellular fields generated by all others. The mathematical form of the resulting coupled system of differential equations mirrors the ancient Pañcikaṛaṇa scheme with striking fidelity. This convergence suggests that both traditions—separated by more than a millennium—recognize the same universal principle: higher-order, macroscopically observable phenomena require the partial “surrender of individuality” of microscopic constituents through precisely weighted mutual interaction.

Organization of the paper

The paper is structured as follows. Section 2 reviews the Vedāntic process of cosmic manifestation as presented in Ādi Śaṅkarācārya’s *Tattva Bodha*. Section 3 presents the biophysical foundation of ephaptic coupling, introduces a representative mathematical model of mutually coupled axons in a tract, and emphasizes that rapid collective synchrony emerges only when every axon receives a weighted contribution from all others—an interaction structure formally identical to Pañcikaṛaṇa. Section 4 draws the detailed parallel between the two frameworks in both mathematical and conceptual terms. Section 5 concludes with a discussion of limitations and future work.

¹See also [3] for the precise scholarly view on the origin of the concept.

Table of Notation

The adjacent table captures the symbols used in this work, along with their meanings and section of appearance.

Table 1: Table of Notation

Symbol	Meaning	Section
e_i ($i = 1, \dots, 5$)	Pure subtle element (tanmātra) corresponding to space, air, fire, water, or earth	1.2
E_i ($i = 1, \dots, 5$)	Gross element (mahābhūta) after Pañcīkaraṇa	1.2
$\frac{1}{2}, \frac{1}{8}$	Proportional weights in the Pañcīkaraṇa mixing process (50% and 12.5%)	1.2
N	Total number of axons in an axonal tract	2
k, j	Indices labelling individual axons ($k, j = 1, 2, \dots, N$)	2
$V_k(t)$	Membrane potential of the k -th axon as a function of time	2
C_m	Membrane capacitance per unit area	2
I_k^{ionic}	Total ionic current of the k -th axon (intrinsic dynamics)	2
g_{kj}	Ephaptic conductance (coupling strength) from axon j to axon k	2
V_{ext}	Reference (extracellular) potential in the coupling term	2

2 The Vedāntic Process of Manifestation

According to Ādi Śaṅkara’s *Tattva Bodha*, cosmic evolution proceeds in four main stages:

2.1 Stage 1 – The Five Subtle Elements (Tanmātras)

From Brahman associated with Māyā arise, sequentially, the five pure subtle elements possessing only their characteristic qualities:

ākāśa (space)	sound
vāyu (air)	sound + touch
agni (fire)	sound + touch + form/color
jala (water)	sound + touch + form + taste
pr̥thivī (earth)	sound + touch + form + taste + smell

At this stage the elements remain unmixed and incapable of producing gross objects.

2.2 Stage 2 – Pañcīkaraṇa (Quintuplication or Grossification)

The inert (*tāmasa*) aspect of each subtle element is divided as follows:

- One half (50%) remains as the “own-substance” of that element.
- The remaining half is subdivided into four equal parts (12.5% each).
- To the intact 50% half of each element is added one 12.5% portion from *each* of the other four elements.

Mathematically, the gross element E_i (where $i = 1, \dots, 5$) is formed as

$$E_i = \frac{1}{2}e_i + \sum_{j \neq i} \frac{1}{8}e_j$$

where $e_1, \dots, e_5$ are the original subtle elements. Thus every gross element contains exactly 50% of itself and 12.5% of every other element.

2.3 Stage 3 – The Five Gross Elements (Mahābhūtas)

The resulting gross elements are fully interdependent and capable of combining to form physical matter.

2.4 Stage 4 – The Gross Physical Body (Sthūla Śarīra)

From these quintuplicated mahābhūtas arises the observable body, subject to the six modifications (birth, existence, growth, maturity, decay, death).

3 Ephaptic Coupling in Axonal Tracts

Ephaptic (non-synaptic, field-mediated) interactions occur when the extracellular electric potential generated by one axon directly influences the membrane potential of neighboring axons. Recent theoretical and experimental work shows that ephaptic coupling is not a pathological curiosity but a normal physiological mechanism enabling sub-millisecond synchrony and collective dynamics in densely packed axonal tracts and cerebellar parallel fibers [1, 2].

Consider a tract of N axons. Let $V_k(t)$ be the membrane potential of the k -th axon. A simplified yet representative model of ephaptic coupling (following Reutskiy, and later Chawla, Morgera, and others) takes the form

$$C_m \frac{dV_k}{dt} = I_k^{\text{ionic}} + \sum_{j=1, j \neq k}^N g_{kj} (V_j(t) - V_{\text{ext}}),$$

where $g_{kj} > 0$ is the ephaptic conductance from axon j to axon k . In anatomically realistic geometries, every axon both generates extracellular fields that affect all others and is itself affected by the weighted sum of fields produced by all others.

Thus, just as an isolated subtle element cannot produce gross matter, an isolated axon cannot participate in tract-wide synchrony or collective wave propagation. Emergent function requires that *each* unit receives a weighted contribution from *every* other unit—precisely the same logical and mathematical structure as Pañcikaraṇa.

4 The Structural and Philosophical Parallel

Vedāntic Pañcikaraṇa	Ephaptic Coupling
Pure tanmātras are functionally inert with respect to gross manifestation	Isolated axonal membranes cannot sustain tract-wide synchrony
Each gross element = 50% self + 12.5% of each other	Each axon's dynamics = own ionic currents + weighted field from <i>every</i> other axon
Weighted mutual contribution is obligatory for higher-order emergence	Weighted mutual coupling is obligatory for collective neural behavior
Without mixing there is no gross body	Without coupling there is no functional neural synchrony

The mathematical isomorphism is not superficial: both frameworks describe systems in which micro-level entities must partially relinquish their independence through fixed proportional contributions to one another in order for macro-level organization to arise.

This convergence is philosophically suggestive. Śaṅkara’s doctrine emphasizes radical interdependence (*anyonya-adhīnatva*) as the condition for manifestation. Contemporary neuroscience increasingly reveals that the brain’s most impressive abilities—ultra-rapid synchronization, gamma coherence, and the binding of distributed activity—similarly depend on the continuous mutual entrainment of millions of neurons via non-synaptic field effects. Isolation, in both traditions, precludes emergence.

5 Conclusion

The parallels of pancikarana drawn in this work might be more widespread than just with ephaptic coupling. They may also be useful as a design principle. So for example when designing a POVM in quantum measurement theory, one may want to do a “coupling” or pancikarana of the component operators. In that way one may see a type of leaked information about other outcomes in each outcome. This investigation is left for future work and holds promise in extending the POVM formalism.

One of the limitations of our study is that it merely uses the Vedantic concept as a “motif” and is not able to probe/study beyond that level in the direction of analysis or extension. This can be addressed in future work as well.

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References

- [1] Han, K. S., et al. (2018). Rapid synchronous activity of cerebellar Purkinje cells during efferent-mediated transmission in the awake mouse. *Nature Communications*, 9(1), 4128.
- [2] Park, H., et al. (2024). Endogenous electric fields shape brain activity through ephaptic coupling. *Nature Reviews Neuroscience*, 25, 112–130.
- [3] P. Deussen, *System of the Vedanta*, Abhishek Publications, 2020.